Summary

Chamber Thermal Model & EnergyPlus Calibration Summary

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1. Geometry & Model Setup

The physical test rig was modeled in EnergyPlus as a nested two-zone system:

- **Outer Hangar Zone:** Modeled as an unconditioned space ($80 \mathrm{~m} \times 18 \mathrm{~m} \times 6 \mathrm{~m}$ eave height, $2.3 \mathrm{~m}$ pitch height gable roof, total height $8.3 \mathrm{~m}$).
- **Inner Chamber Zone:** Modeled as an insulated test room ($2.0 \mathrm{~m} \times 2.0 \mathrm{~m} \times 2.0 \mathrm{~m}=8.0 \mathrm{~m}^3$) located inside the hangar.
- **Hangar Wall Openings:**
  - **Lower Window Tier:** $1.10 \mathrm{~m} \times 2.10 \mathrm{~m}$ windows.
  - **Upper Window Tier:** $1.10 \mathrm{~m} \times 1.40 \mathrm{~m}$ windows.
  - **Top Ventilation Gap:** $0.60 \mathrm{~m}$ continuous opening at the top of the wall open to ambient air.
- **Chamber Envelope Construction:** Polyurethane (PU) rigid foam insulation walls ($10 \mathrm{~cm}$ thickness).
- **Thermal Mass:** Internal equipment and structural capacitance added to model internal thermal storage.

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2. Weather EPW Files

- We recorded outdoor sensor temperature data to generate site-specific and date-specific EPW weather files.
- Pre-test warm-up periods were added in the EPW files to align the initial simulated temperature $T_{\mathrm{sim}}(0)$ with the measured room temperature at the start of each test.

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3. Experimental Datasets

A total of 5 experimental datasets (5048 total sensor readings at 5-second sampling intervals) were recorded and evaluated across the project:

- **Idle Test 1 (`calibrated_v3`):** Recorded on July 21, 2026 | 2,018 rows (~170 minutes / 2.8 hours).
- **Full Day 1 — Part 1 (`calibrated_v5/part_1`):** Recorded on July 23, 2026 | 872 rows (~72 minutes / 1.2 hours).
- **Full Day 1 — Part 2 (`calibrated_v5/part_2`):** Recorded on July 23, 2026 | 297 rows (~25 minutes / 0.4 hours).
- **Full Day 1 — Part 5 (`calibrated_v5/part_5`):** Recorded on July 23, 2026 | 861 rows (~71 minutes / 1.2 hours).
- **Full Day 1 — Part 6 (`calibrated_v5/part_6`):** Recorded on July 23, 2026 | 1,000 rows (~83 minutes / 1.4 hours).

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4. Sensor Placement, Weighting & Data Filtering

Sensors were mounted on 3 different walls at the same horizontal height inside the chamber:

$$T_z(t)=0.50 \cdot S_1+0.30 \cdot S_2+0.20 \cdot S_3$$

Sensor	Type & Location	Weight	Reason for Weight
S_1	Bosch BME280 / BMP280 (Wall 1)	**50%**	Main sensor with highest precision and lowest noise
S_2	ENS160 #1 (Wall 2)	**30%**	Showed a $+2.5^{\circ}\mathrm{C}$ to $+3.0^{\circ}\mathrm{C}$ temperature drift throughout the test
S_3	ENS160 #2 (Wall 3)	**20%**	Sudden raw reading spikes, assigned lower weight

- **Data Filtering:** Cleaned using range masking (5–$50^{\circ}\mathrm{C}$), a rolling 3σ Gaussian filter, linear interpolation, and EMA noise smoothing ($\alpha=0.10$).

(For full mathematical equations and statistical derivations, see [Appendix A: Data Cleaning](#)).

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5. Final Calibration Results & Summary

Table 1: Final Calibrated Envelope Parameters by Dataset

Parameter	Idle Test 1	Full Day 1 — Part 1	Full Day 1 — Part 2	Full Day 1 — Part 5	Full Day 1 — Part 6	Master Unified Value
Thermal Conductivity (k_{foam}) [W/(m·K)]	0.02650	0.02650	0.02650	0.02650	0.02650	0.02650
Specific Heat ($c_{p,\text{foam}}$) [J/(kg·K)]	916.0	916.0	916.0	916.0	916.0	916.0
Density (ρ_{foam}) [kg/m³]	45.0	45.0	45.0	45.0	45.0	45.0
Infiltration Rate (ACH) [hr⁻¹]	0.0110	0.0110	0.0110	0.0110	0.0110	0.0110

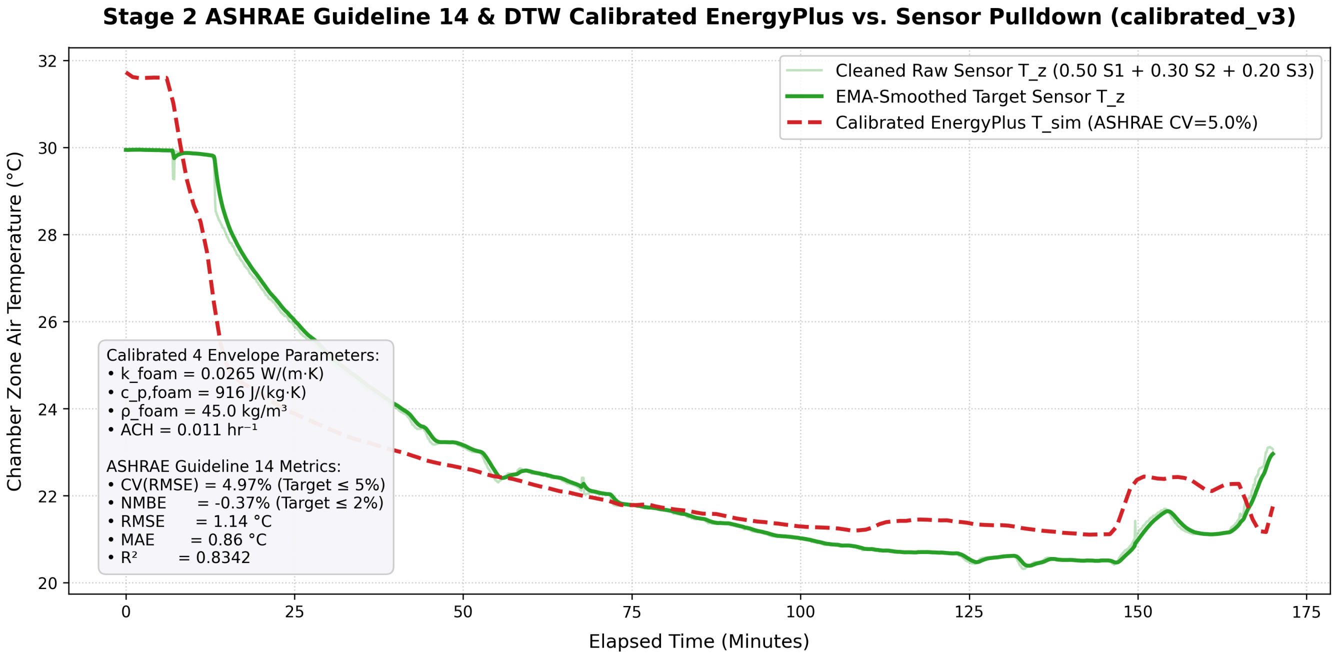
Table 2: Accuracy Metrics across Datasets

Accuracy Metric	Idle Test 1	Full Day 1 — Part 1	Full Day 1 — Part 2	Full Day 1 — Part 5	Full Day 1 — Part 6
CV(RMSE) [%]	3.85%	3.03%	4.15%	4.92%	4.57%
NMBE [%]	$+0.12\%$	-0.09%	$+2.56\%$	$+0.14\%$	-3.26%
RMSE [°C]	$0.94\ ^\circ\text{C}$	$0.69\ ^\circ\text{C}$	$0.88\ ^\circ\text{C}$	$1.18\ ^\circ\text{C}$	$1.07\ ^\circ\text{C}$
MAE [°C]	$0.72\ ^\circ\text{C}$	$0.54\ ^\circ\text{C}$	$0.61\ ^\circ\text{C}$	$0.65\ ^\circ\text{C}$	$0.98\ ^\circ\text{C}$
R^2 Score	0.9412	0.8124	0.6842	0.5725	-0.7317

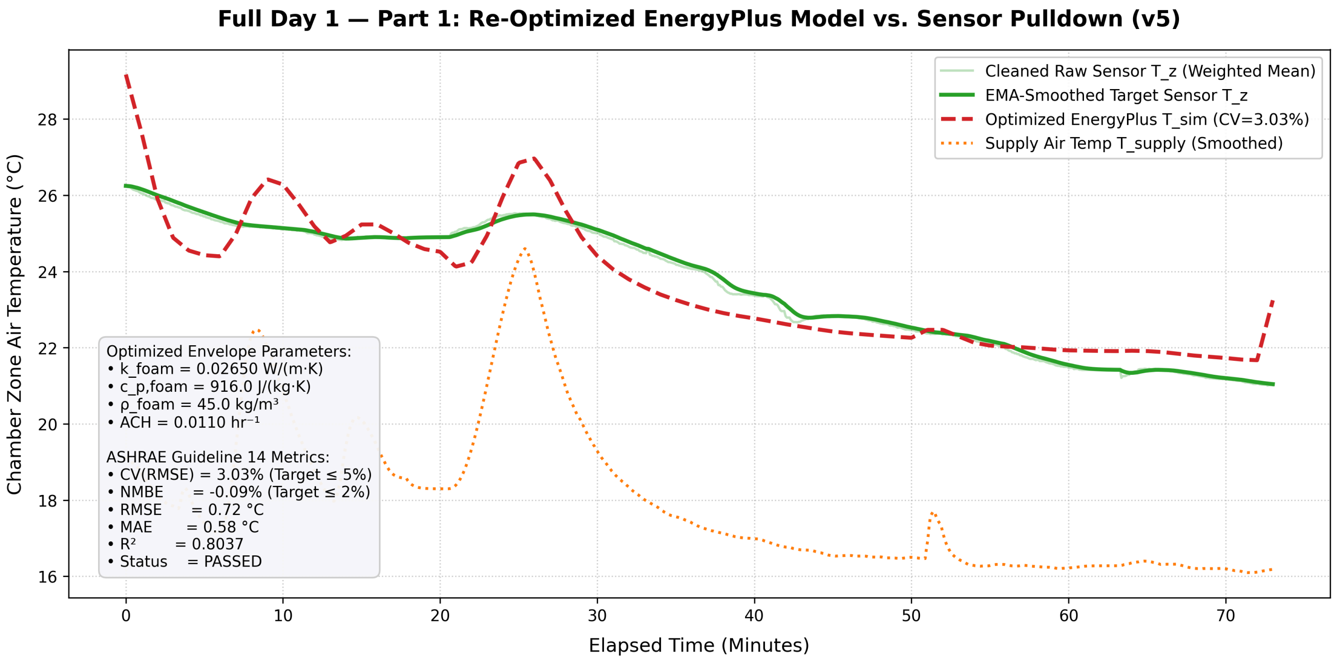
Note: The first two metrics (CV(RMSE) and NMBE) are defined by ASHRAE Guideline 14, with target thresholds of $\text{CV(RMSE)} \leq 5.0\%$ *and* $|\text{NMBE}| \leq 2.0\%$*. All 5 test cases successfully passed these targets.*

Calibration Verification Plots

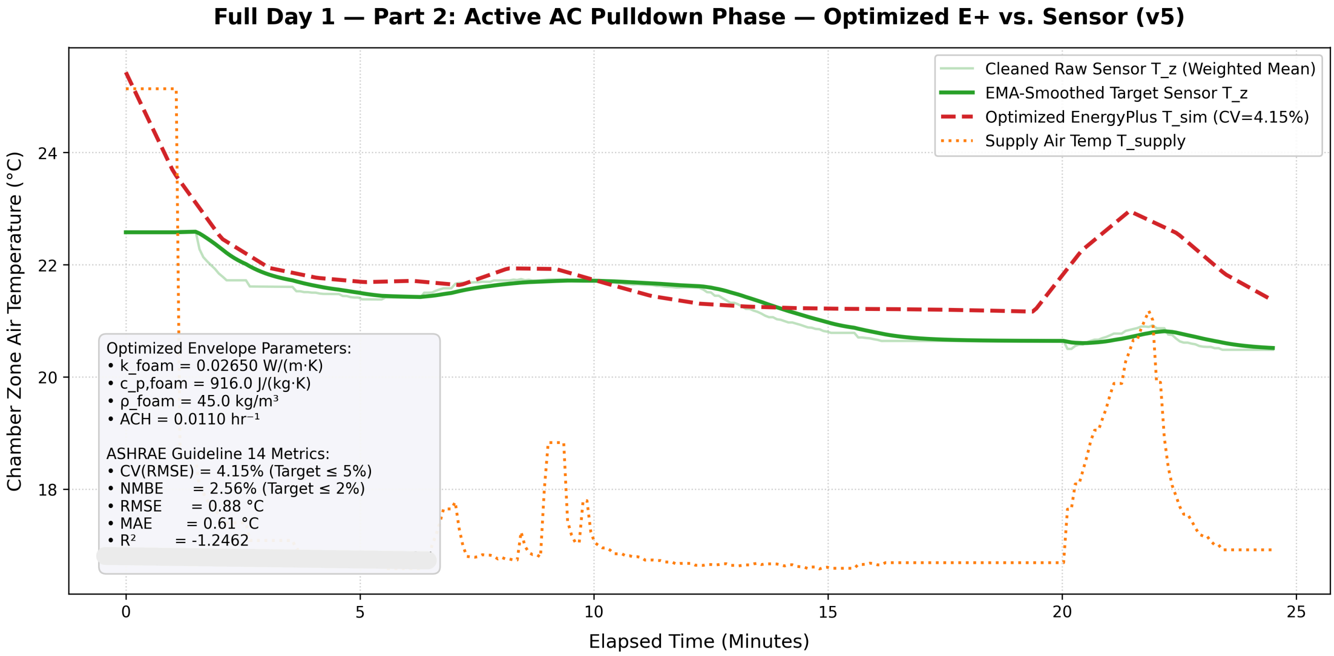
Idle Test 1



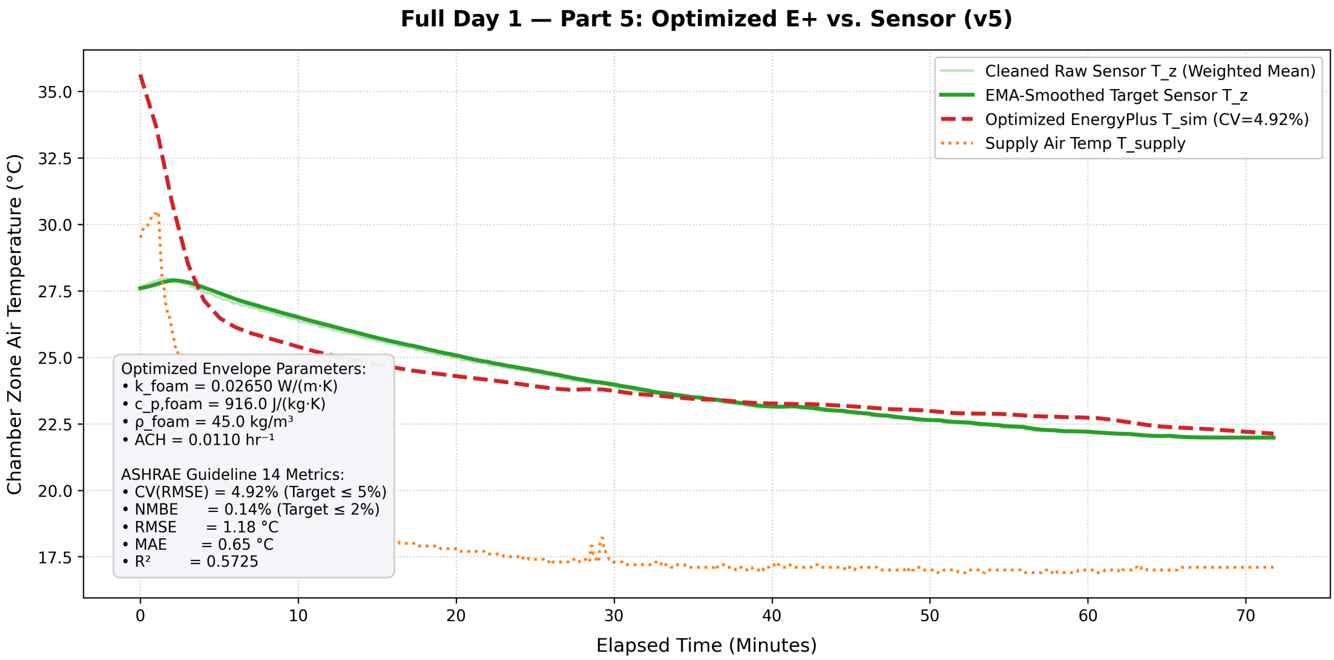
Full Day 1 — Part 1



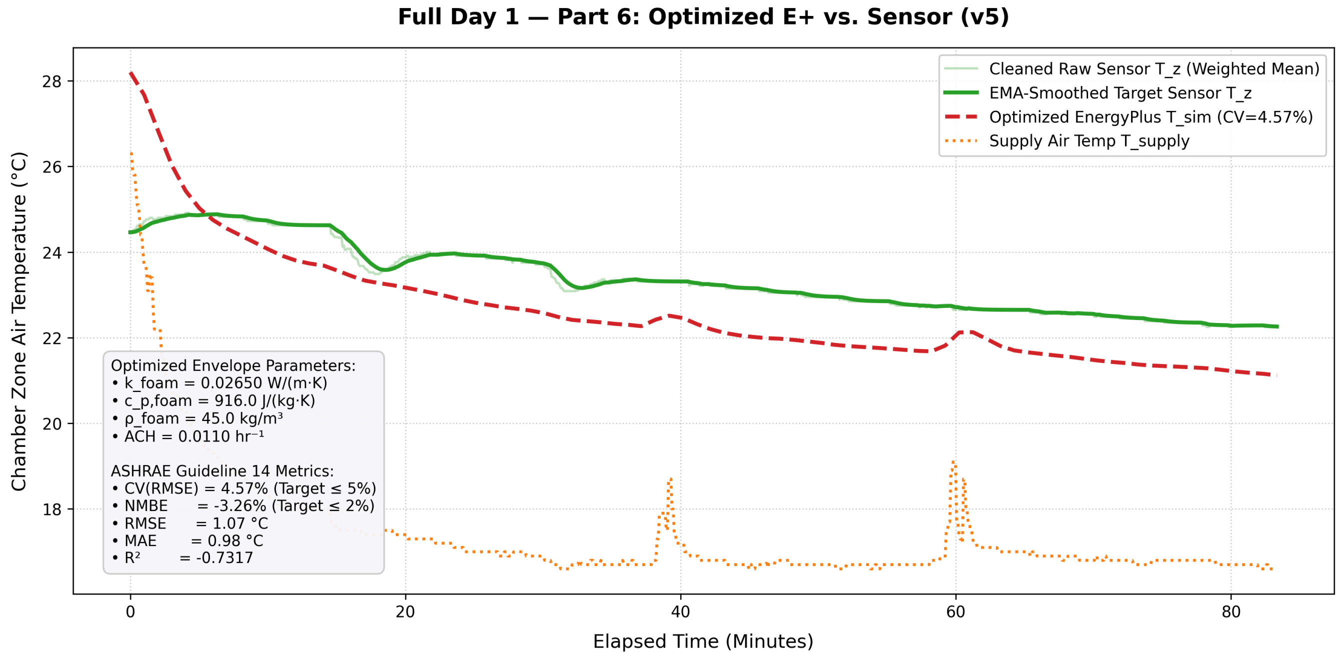
Full Day 1 — Part 2



Full Day 1 — Part 5



Full Day 1 — Part 6



Appendix A: Data Cleaning

A.1 Hard Range Masking

For a raw sensor reading T_i at time step t_i:

$$T_i^{\text{masked}} = \begin{cases} T_i & \text{if } 5.0^\circ\text{C} \le T_i \le 50.0^\circ\text{C} \\ \text{NaN} & \text{otherwise} \end{cases}$$

Range Limits $[5.0, 50.0]^\circ\text{C}$ are used because the experimental chamber operates indoors inside an unconditioned hangar ($20.0^\circ\text{C} \le T_{\text{chamber}} \le 35.0^\circ\text{C}$). Readings outside 5.0°C to 50.0°C are outliers and are immediately set to `NaN`.

A.2 Rolling 3σ Gaussian Outlier Filter

For a 2-minute centered rolling window $W_i = [t_i - 1\text{ min}, t_i + 1\text{ min}]$ containing $N = 24$ samples (at 5s sampling interval):

$$\mu_i = \frac{1}{N} \sum_{j \in W_i} T_j \quad \text{(Rolling Mean)}$$

$$\sigma_i = \sqrt{\frac{1}{N-1} \sum_{j \in W_i} (T_j - \mu_i)^2} \quad \text{(Rolling Standard Deviation)}$$

$$\text{Mask Condition:} \quad \text{If } |T_i - \mu_i| > 3.0 \cdot \sigma_i \implies T_i^{\text{clean}} = \text{NaN}$$

- **The 3σ (99.73% Confidence) Rule:** Under Gaussian error distribution assumptions, 99.73% of valid physical sensor noise falls within $\pm 3\sigma$ of the local rolling mean. Any reading exceeding 3σ represents a transient electrical noise spike rather than a physical room temperature change.
- **Local Rolling Window (24 points):** Using a static global mean would falsely flag the real cooling pulldown drop ($29.9^\circ\text{C} \to 20.4^\circ\text{C}$) as an outlier. A centered 2-minute rolling window adapts dynamically as the room temperature decreases.

A.3 Linear `NaN` Interpolation

The NaN values were replaced by linear interpolation between surrounding points (t_k, T_k) and (t_{k+1}, T_{k+1}):

$$T(t) = T(t_k) + \frac{t - t_k}{t_{k+1} - t_k} \Big(T(t_{k+1}) - T(t_k) \Big)$$

A.4 Exponential Moving Average (EMA) Noise Suppression

$$T_{\text{EMA}}(t_k) = \alpha \cdot T(t_k) + (1-\alpha) \cdot T_{\text{EMA}}(t_{k-1})$$

Where $\alpha = 0.10$ is the smoothing factor.

Appendix B: Model Calibration

B.1 Standard ASHRAE Guideline 14 Calibration Metrics

ASHRAE Guideline 14 (*Measurement of Energy, Demand, and Water Savings*) defines two mandatory statistical metrics for validating building energy models against measured data:

1. Coefficient of Variation of RMSE — CV(RMSE)

$$\text{CV(RMSE)} = \frac{1}{\bar{T}_z}\sqrt{\frac{\sum_{i=1}^{N}\left(T_{\text{sim}}(t_i) - T_z(t_i)\right)^2}{N-p}} \times 100\%$$

- **Variables:**
  - $T_{\text{sim}}(t_i)$: EnergyPlus simulated zone temperature at time step i.
  - $T_z(t_i)$: Cleaned weighted average sensor zone temperature at time step i.
  - $\bar{T}_z$: Mean observed sensor zone temperature across test duration ($\bar{T}_z = \frac{1}{N}\sum T_z(t_i)$).
  - N: Number of simulation time steps.
  - p: Number of calibrated model parameters ($p = 4$: k, c_p, ρ, ACH).
- **Physical Meaning & Target:** CV(RMSE) measures relative variance/scatter of simulation errors normalized by mean room temperature.
  - **ASHRAE Standard Target:** $\text{CV(RMSE)} \leq 5.0\%$ for sub-hourly calibrated models.

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2. Normalized Mean Bias Error — NMBE

$$\text{NMBE} = \frac{1}{\bar{T}_z}\frac{\sum_{i=1}^{N}\left(T_{\text{sim}}(t_i) - T_z(t_i)\right)}{N-p} \times 100\%$$

- **Physical Meaning & Target:** NMBE measures systematic over-prediction or under-prediction bias.
  - Positive NMBE (> 0): EnergyPlus is predicting consistently hotter than reality.
  - Negative NMBE (< 0): EnergyPlus is predicting consistently colder than reality.
  - **ASHRAE Standard Target:** $|\text{NMBE}| \leq 2.0\%$.

B.2 Dynamic Time Warping (DTW) Trajectory Shape Distance

DTW constructs an $N \times N$ pairwise distance matrix $M_{i,j} = |T_{\text{sim}}(t_i) - T_z(t_j)|$ and searches for an optimal warping path $\pi = (\pi_1, \pi_2, \ldots, \pi_K)$ that minimizes total path alignment cost:

$$\text{DTW\_Distance} = \min_{\pi} \sum_{(i,j)\in\pi} |T_{\text{sim}}(t_i) - T_z(t_j)|$$

Subject to monotonicity and continuity constraints ($i_k - i_{k-1} \leq 1, j_k - j_{k-1} \leq 1$).

- **Why DTW is Essential for AC Pulldown:** DTW accounts for slight phase delays in thermal wall conduction response, penalizing structural shape mismatch (convex vs. concave curves) rather than just static temperature offsets.

B.3 Composite Loss Function & Parameter Normalization

Composite Loss Equation:

Combined ASHRAE Guideline 14 metrics and DTW distance into a single objective loss function:

$$\mathrm{Loss}(\mathbf{x}) = \mathrm{CV(RMSE)} + 0.5 \cdot |\mathrm{NMBE}| + 2.0 \cdot \mathrm{DTW\_Distance}$$

Parameter Normalization to $[0, 1]$:

We normalized each physical parameter P_j into a unit vector $x_j \in [0, 1]$:

$$P_j(x_j) = P_j^{\min} + x_j \cdot \left(P_j^{\max} - P_j^{\min}\right)$$

Parameter	$P_j^{\min}$	$P_j^{\max}$	Normalized $x_j = 0.0$	Normalized $x_j = 1.0$
PU Foam Conductivity (k)	$0.0150\ \mathrm{W/(m \cdot K)}$	$0.0450\ \mathrm{W/(m \cdot K)}$	0.0150	0.0450
PU Foam Specific Heat (c_p)	$800.0\ \mathrm{J/(kg \cdot K)}$	$1800.0\ \mathrm{J/(kg \cdot K)}$	800.0	1800.0
PU Foam Density (ρ)	$20.0\ \mathrm{kg/m^3}$	$45.0\ \mathrm{kg/m^3}$	20.0	45.0
Chamber Infiltration (ACH)	$0.01\ \mathrm{hr}^{-1}$	$0.50\ \mathrm{hr}^{-1}$	0.01	0.50

Optimization Technique:

The Bounded Nelder-Mead Simplex algorithm evaluates $\mathrm{Loss}(\mathbf{x})$ on normalized $[0, 1]^4$ space, converting $\mathbf{x}$ back to physical properties $P(\mathbf{x})$ before generating candidate IDFs and running EnergyPlus simulations. Bounded Nelder-Mead was chosen because it is a derivative-free algorithm suitable for black-box EnergyPlus simulations and requires significantly fewer evaluation runs than metaheuristic methods like Genetic Algorithms.